Experiment Number: 04 **Date: 05/02/2026**

ARRAYS

Aim:

To understand and implement array operations in Java.

PRE LAB EXERCISE

QUESTIONS

- ✓ **What is an array?**

  It is a collection of data itmes of same data type.

- ✓ **Why are arrays used?**

  ->Store multiple values under single name.

  ->It makes easy access and faster execution.

- ✓ **What is the difference between array and variable?**

Variable	**Array**
Stores only one value	**Stores multiple values**
Needs separate variables for each value	**Uses one name for many values**
Example: int a = 10;	**Example: int a[] = {10, 20, 30};**

IN LAB EXERCISE

Objective:

To perform array operations using simple programs.

PROGRAMS:

1. Program to Read and Print Array Elements

Code:

```
import java.util.Scanner;
```

```java
public class ReadPrintArray {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int[] arr = new int[5];
    System.out.println("Enter 5 elements:");
    for(int i = 0; i < 5; i++)
      arr[i] = sc.nextInt();
    System.out.println("Array elements are:");
    for(int i = 0; i < 5; i++)
      System.out.print(arr[i] + " ");
  }
}
```

OUTPUT:

Input:
10 20 30 40 50

Output:

Array elements are:

10 20 30 40 50

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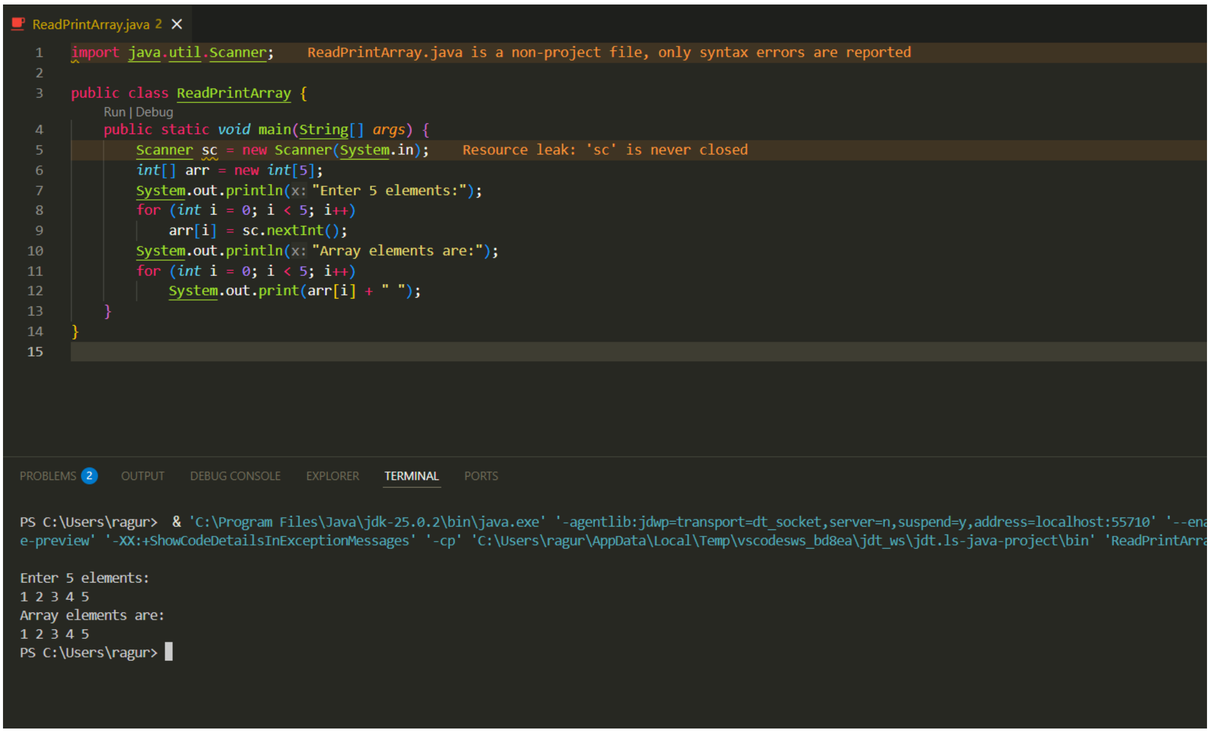


2. Program to Find Sum of Array Elements

Code:

```
import java.util.Scanner;
public class SumArray {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int[] arr = new int[5];
    int sum = 0;
    System.out.println("Enter 5 elements:");
    for(int i = 0; i < 5; i++)
      arr[i] = sc.nextInt();
    for(int i = 0; i < 5; i++)
      sum += arr[i];
    System.out.println("Sum = " + sum);
  }
```

```
}
```

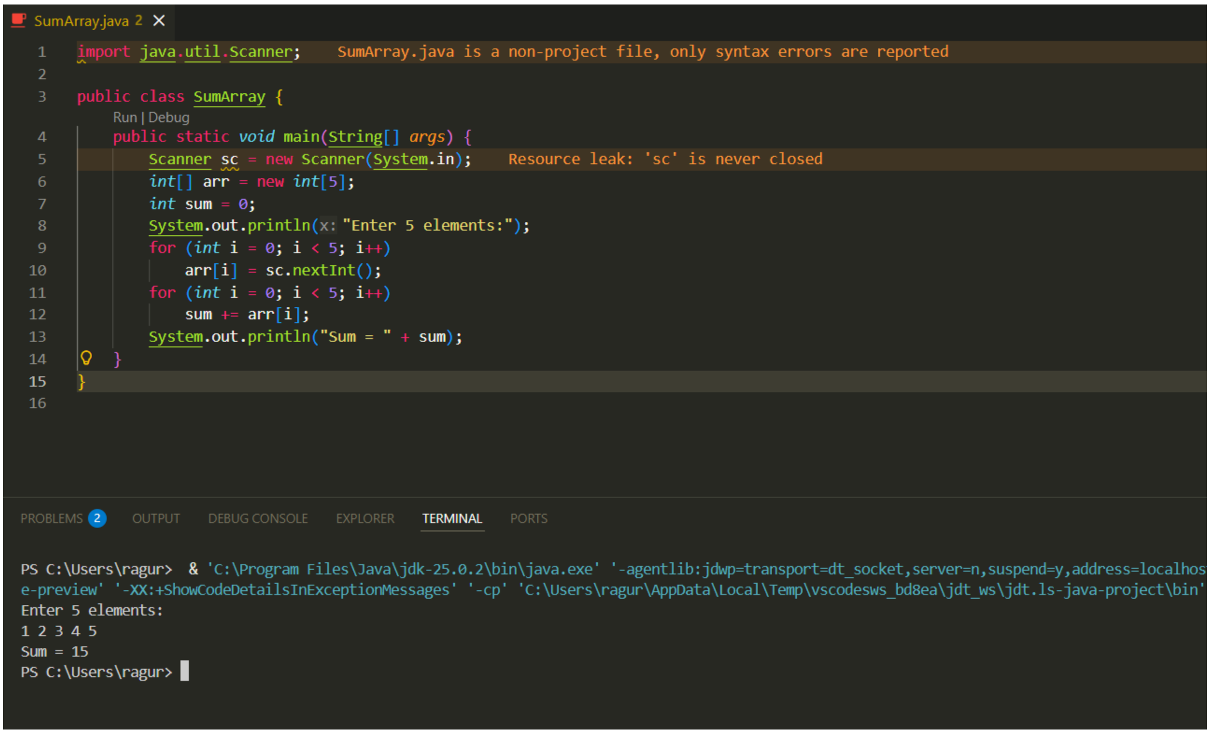
OUTPUT:

Input:

5 10 15 20 25

Output:

Sum = 75



3. Program to Find Largest Element in an Array

Code:

```
import java.util.Scanner;

public class LargestElement {

  public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    int[] arr = new int[5];

    System.out.println("Enter 5 elements:");

    for(int i = 0; i < 5; i++)

      arr[i] = sc.nextInt();

    int max = arr[0];
```

```
        for(int i = 1; i < 5; i++)
            if(arr[i] > max)
                max = arr[i];
        System.out.println("Largest element = " + max);
    }
}
```

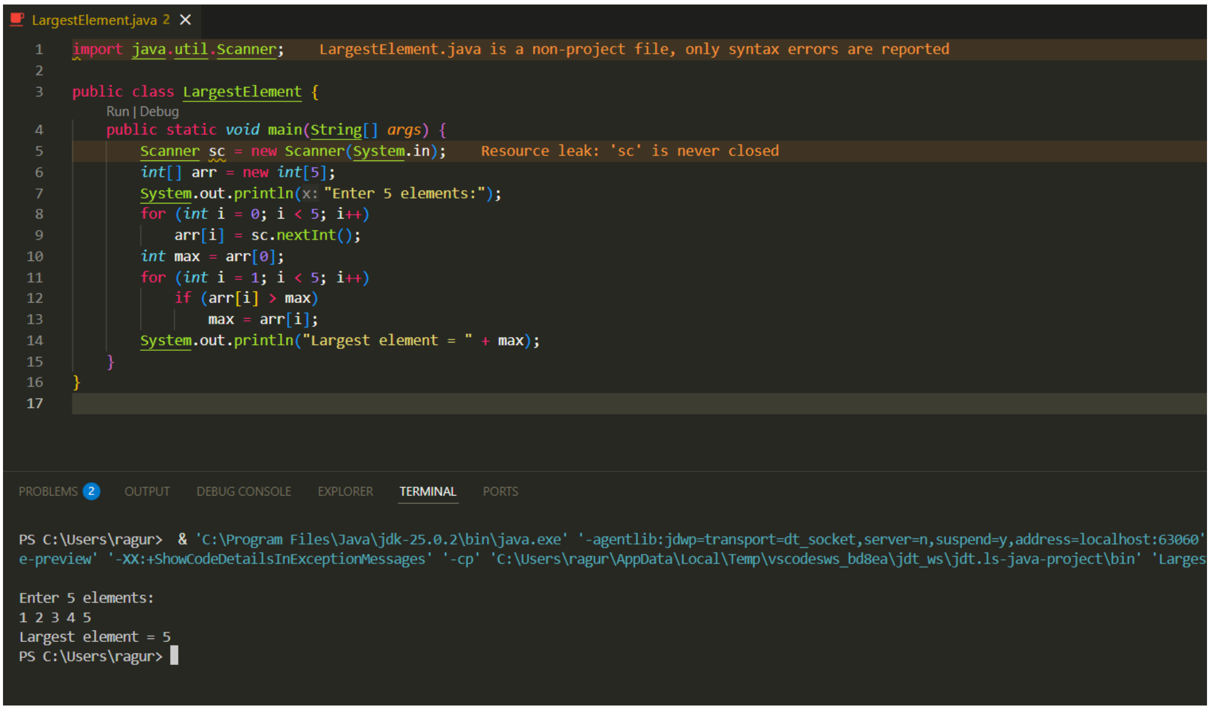
OUTPUT:

Input:

12 45 23 9 30

Output:

Largest element = 45



4. Program to Reverse an Array

Code:

```
import java.util.Scanner;

public class ReverseArray {
```

```
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int[] arr = new int[5];
    System.out.println("Enter 5 elements:");
    for(int i = 0; i < 5; i++)
        arr[i] = sc.nextInt();
    System.out.println("Reversed array:");
    for(int i = 4; i >= 0; i--)
        System.out.print(arr[i] + " ");
  }
}
```

OUTPUT:

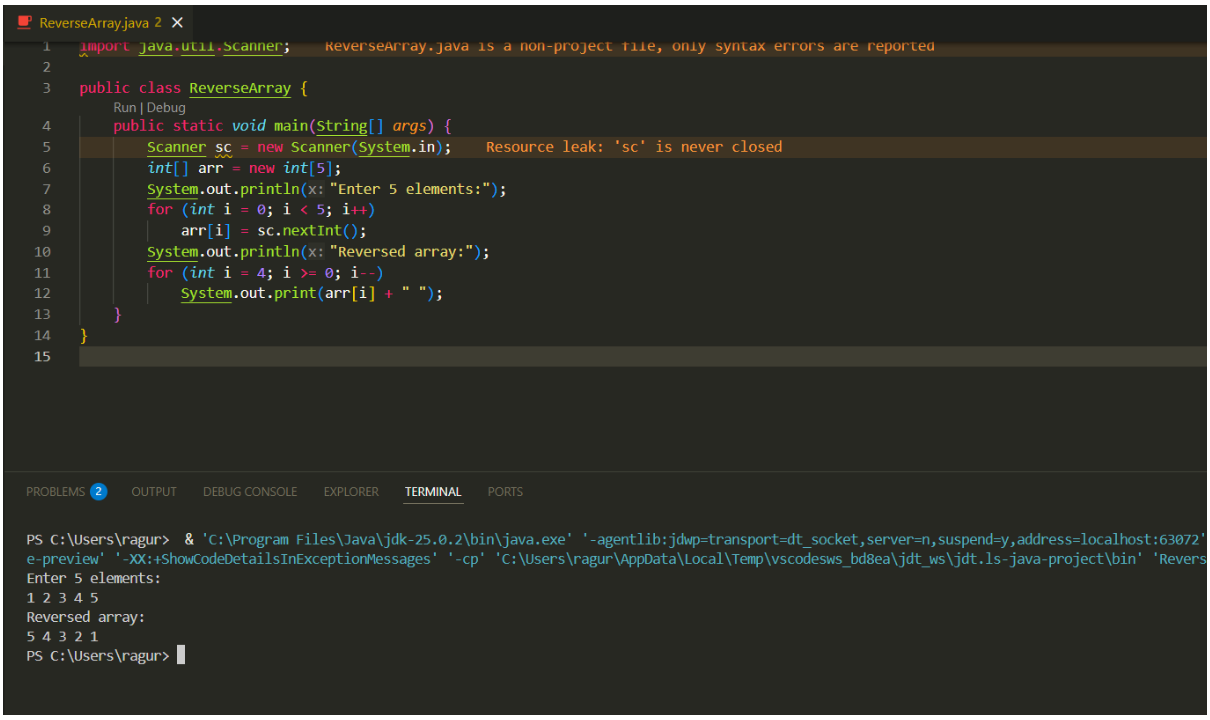
Input:

1 2 3 4 5

Output:

Reversed array:

5 4 3 2 1



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5. Program to Count Even and Odd Numbers

Code:

```
import java.util.Scanner;
public class EvenOddCount {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int[] arr = new int[5];
    int even = 0, odd = 0;
    System.out.println("Enter 5 elements:");
    for(int i = 0; i < 5; i++)
      arr[i] = sc.nextInt();
    for(int i = 0; i < 5; i++) {
      if(arr[i] % 2 == 0)
        even++;
      else
        odd++;
    }

    System.out.println("Even = " + even);
    System.out.println("Odd = " + odd);
  }
}
```

OUTPUT:

Input:
2 7 4 9 10

Output:

Even = 3

Odd = 2

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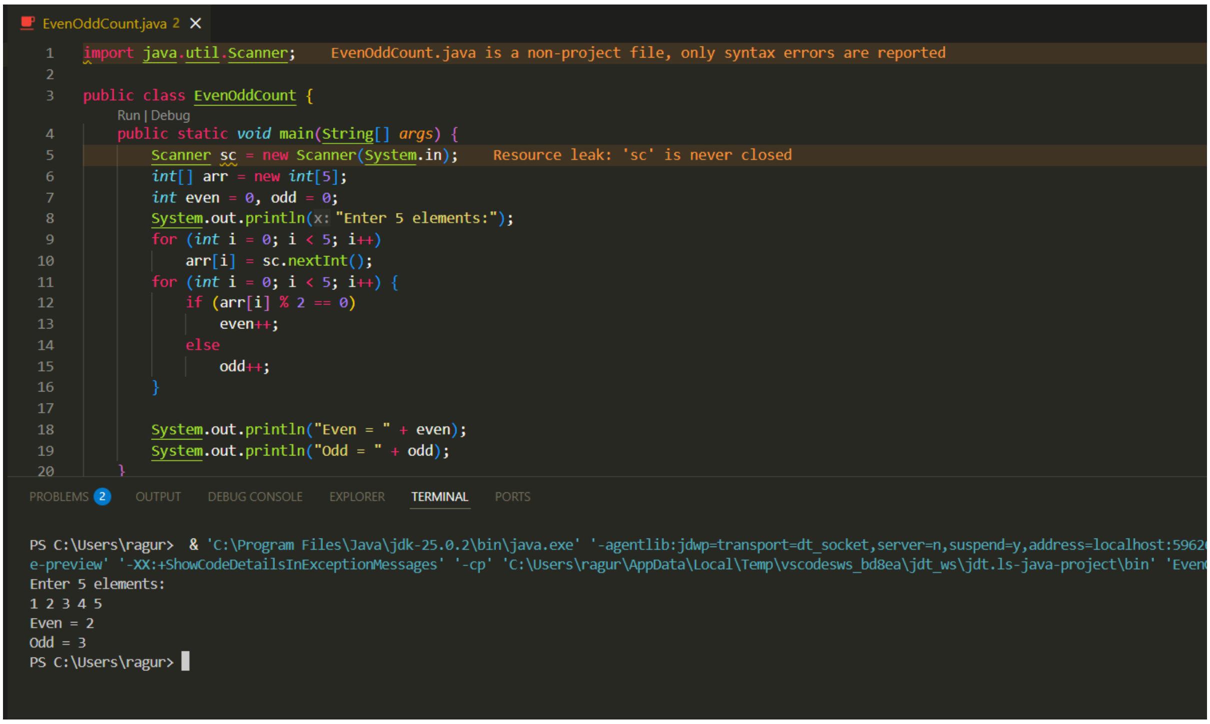


6. Program to Sort Array in Ascending Order

Code:

```
import java.util.Scanner;

public class SortArray {

  public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    int[] arr = new int[5];

    int temp;

    System.out.println("Enter 5 elements:");

    for(int i = 0; i < 5; i++)

      arr[i] = sc.nextInt();

    for(int i = 0; i < 5; i++) {

      for(int j = i + 1; j < 5; j++) {

        if(arr[i] > arr[j]) {

          temp = arr[i];

          arr[i] = arr[j];
```

```
            arr[j] = temp;
          }
        }
      }
      System.out.println("Sorted array:");
      for(int i = 0; i < 5; i++)
        System.out.print(arr[i] + " ");
    }
}
```

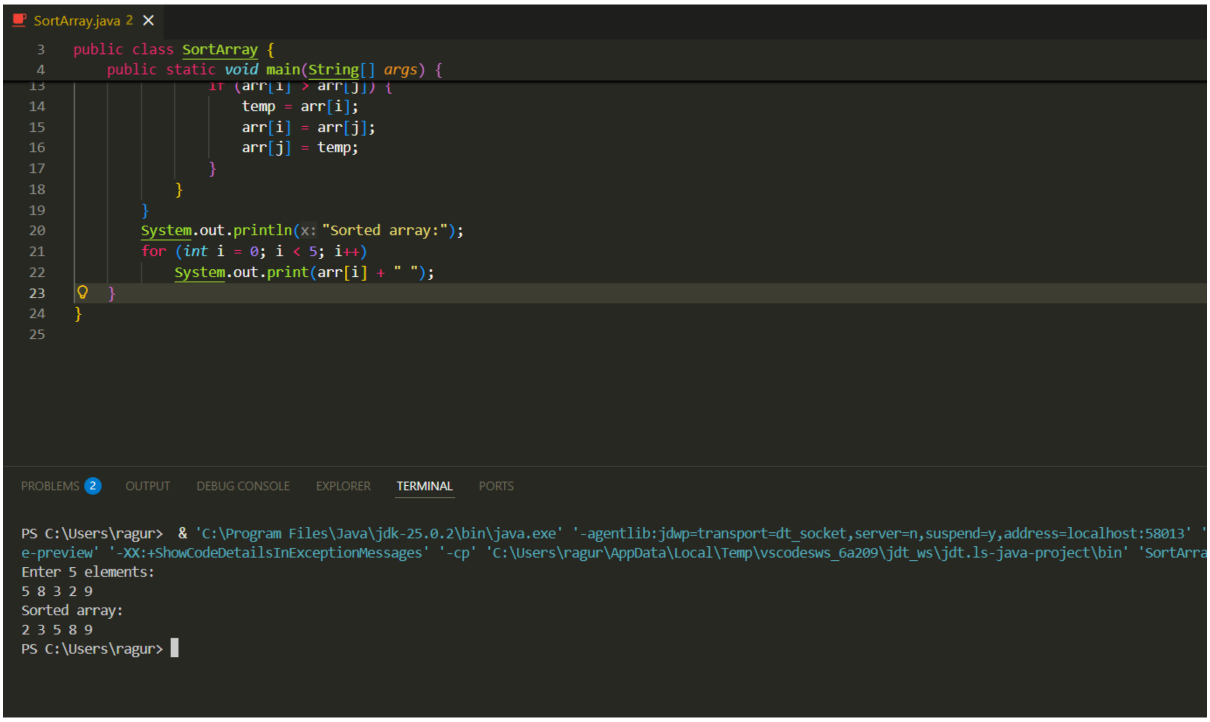
OUTPUT:

Input:
45 12 78 23 9

Output:

Sorted array:

9 12 23 45 78



7. Program to Find Second Largest Element

Code:

```
import java.util.Scanner;
public class SecondLargest {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int[] arr = new int[5];

    System.out.println("Enter 5 elements:");
    for(int i = 0; i < 5; i++)
      arr[i] = sc.nextInt();
    int largest = arr[0];
    int second = arr[0];
    for(int i = 0; i < 5; i++) {
      if(arr[i] > largest) {
        second = largest;
        largest = arr[i];
      }
    }
    System.out.println("Second largest = " + second);
  }
}
```

OUTPUT:

Input:
10 45 23 89 67

Output:

Second largest = 67

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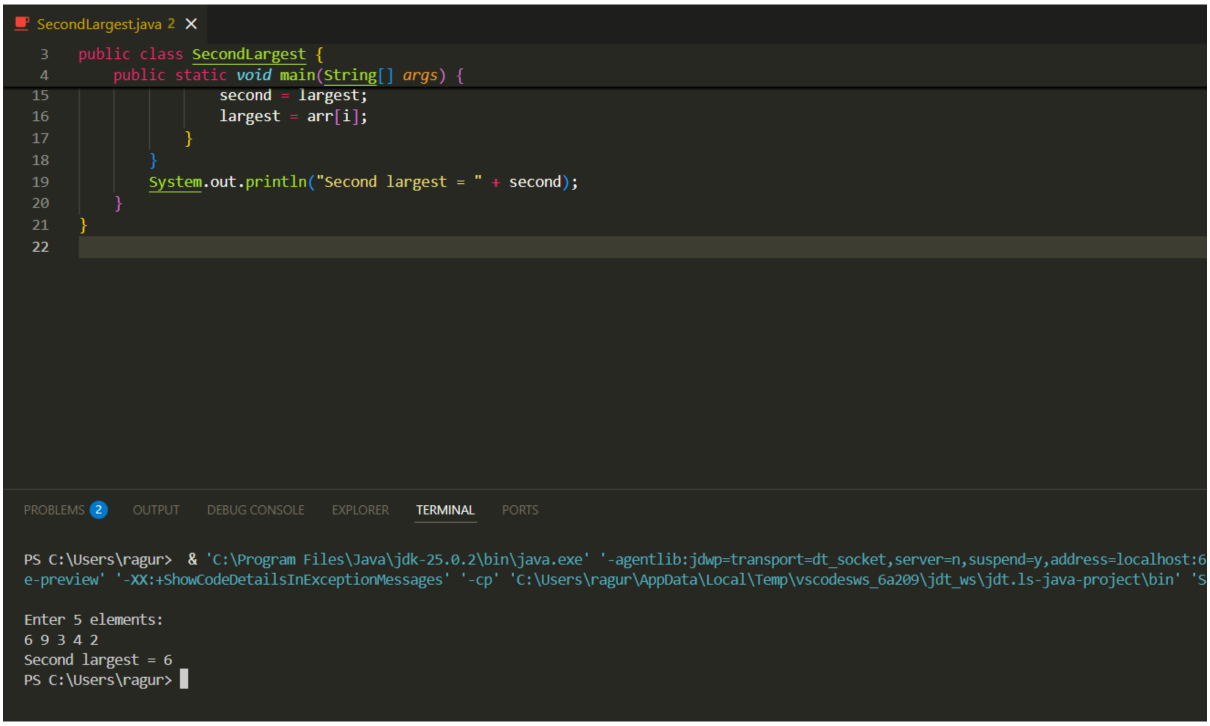

8. Program for Matrix Addition (2D Array)

Code:

```
import java.util.Scanner;

public class MatrixAddition {

  public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    int[][] a = new int[2][2];

    int[][] b = new int[2][2];

    int[][] sum = new int[2][2];

    System.out.println("Enter elements of matrix A:");

    for(int i = 0; i < 2; i++)

      for(int j = 0; j < 2; j++)

        a[i][j] = sc.nextInt();

    System.out.println("Enter elements of matrix B:");

    for(int i = 0; i < 2; i++)
```

```
        for(int j = 0; j < 2; j++)
            b[i][j] = sc.nextInt();
    for(int i = 0; i < 2; i++)
        for(int j = 0; j < 2; j++)
            sum[i][j] = a[i][j] + b[i][j];
    System.out.println("Sum matrix:");
    for(int i = 0; i < 2; i++) {
        for(int j = 0; j < 2; j++)
            System.out.print(sum[i][j] + " ");
        System.out.println();
    }
  }
}
```

OUTPUT:

Matrix A:

1 2

3 4

Matrix B:

5 6

7 8

Sum matrix:

6 8

10 12

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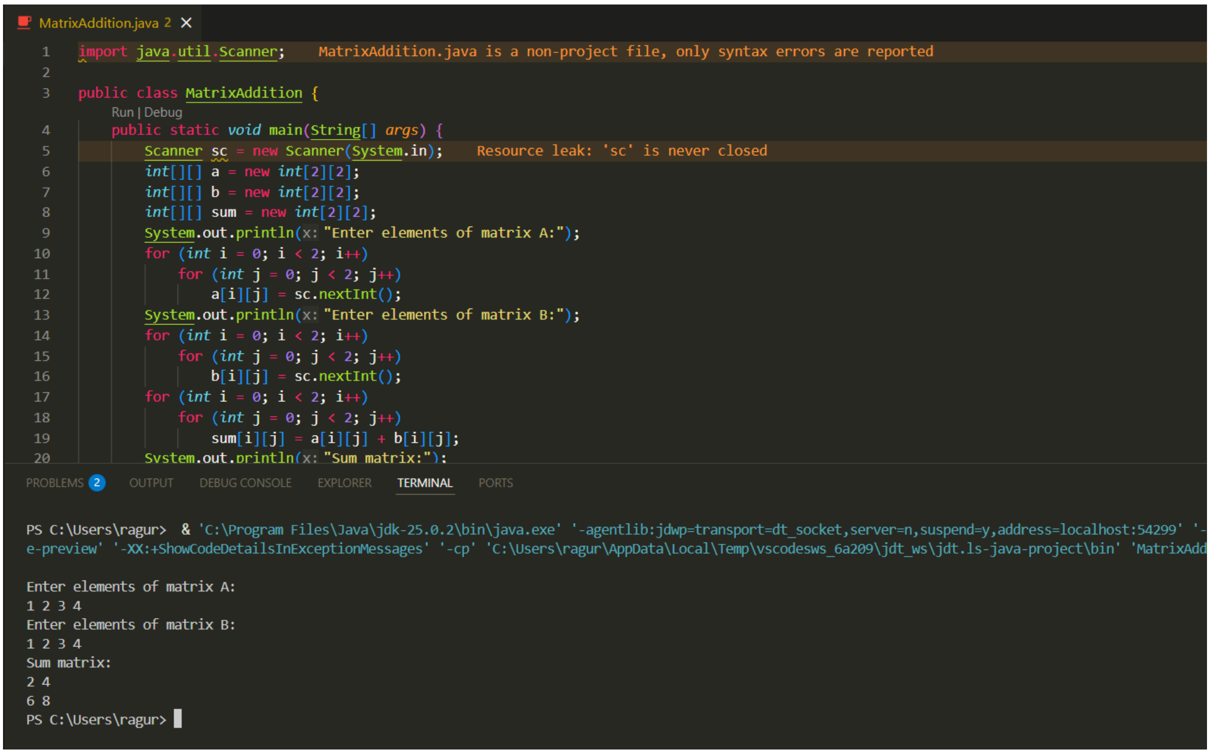

POST LAB EXERCISE

- ✓ **Why is array indexing usually started from zero instead of one?**

  Because index 0 means the first memory location, making access faster and easier.

- ✓ **What happens if we try to access an array element outside its declared size?**

  It causes an error.

- **Java → ArrayIndexOutOfBoundsException**

- ✓ **How does memory allocation differ for static arrays and dynamic arrays?**

  - **Static array**: Size is fixed, decided early
  - **Dynamic array**: Size can change during program execution

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✓ **Why is searching faster in arrays compared to linked lists?**

Because arrays allow direct access using index, but linked lists do not.

✓ **What is the difference between contiguous and non-contiguous memory allocation?**

- **Contiguous**: Stored together → faster (arrays)
- **Non-contiguous**: Stored separately → slower (linked lists)

Result:

Thus the array operations were executed successfully.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		

Ranjith B
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